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Subject: Mobile Application Development

CA- Mini Project

Project Title: Employee Attendance Tracker

Introduction:

The Employee Attendance Tracker is a mobile application developed using Flutter and Dart that helps organizations, schools, offices, and small businesses manage employee attendance digitally. The application replaces the traditional paper-based attendance register with an efficient digital system.

In many organizations, attendance is still maintained manually using registers or spreadsheets. This process is time-consuming, prone to human errors, and makes it difficult to generate reports or monitor employee attendance. The Employee Attendance Tracker solves these problems by allowing administrators to maintain employee records, mark daily attendance, and generate attendance reports through an easy-to-use mobile application.

The application is designed to work completely offline using SQLite, a lightweight local database. Since all data is stored on the mobile device, there is no need for an internet connection or cloud server, making the application suitable for small businesses and organizations.

The application provides an intuitive user interface where administrators can easily add employees, update employee information, record attendance, search employees, and view attendance reports. The system is simple enough for beginners while demonstrating professional mobile application development concepts.

Background:

Attendance management is an essential task in every organization because it directly affects employee performance monitoring, payroll preparation, and overall workforce management.

Traditionally, attendance is maintained using paper registers or spreadsheets. Although these methods are simple, they become inefficient as the number of employees increases. Manual attendance systems consume time, require additional paperwork, and make report generation difficult.

With the increasing use of smartphones, mobile applications provide an efficient alternative for attendance management. A mobile attendance application enables

managers to record attendance instantly, store data securely, and generate reports within seconds.

The Employee Attendance Tracker has been developed to provide a simple, offline, and user-friendly solution for attendance management using Flutter and SQLite.

Need for the Project:

Many small businesses, coaching institutes, startups, and local offices cannot afford expensive Human Resource Management Systems (HRMS). They require a simple attendance management application that works offline without requiring cloud services or subscription costs.

This project fulfils that requirement by providing:

- A low-cost attendance management solution.
- Easy employee record management.
- Fast attendance marking.
- Automatic attendance report generation.

Importance of the Project:

Employee attendance is one of the most important aspects of workforce management. Accurate attendance records help organizations:

- Monitor employee punctuality.
- Improve workforce productivity.
- Reduce manual errors.
- Save administrative time.
- Generate attendance reports quickly.
- Maintain proper employee records.
- Support salary and leave calculations in future versions.

Benefits of the Project:

- Reduces paperwork.
- Easy attendance recording.
- Faster report generation.
- User-friendly interface.
- Offline functionality.
- Secure local data storage.
- Low maintenance cost.
- Saves time and effort.
- Easy to use for non-technical users.
- Portable and accessible on Android devices.

Proposed System:

- The proposed system is an Employee Attendance Tracker mobile application developed using Flutter and SQLite.
- The application provides an easy way to manage employee records and attendance digitally. It allows administrators to add employees, update information, mark daily attendance, search employee records, and generate attendance reports.
- All information is stored locally using SQLite, allowing the application to function completely offline without requiring an internet connection.
- The proposed system is designed to be simple, fast, and suitable for small businesses and organizations.

Objectives:

The main objectives of the project are:

- Develop a simple mobile attendance system.
- Digitally manage employee information.
- Record daily attendance efficiently.
- Reduce paperwork.
- Generate attendance summaries.
- Provide quick employee search.
- Store attendance securely using SQLite.
- Build an easy-to-use interface.
- Work completely offline.

Scope of the Project:

The scope of the Employee Attendance Tracker includes:

Employee Management:

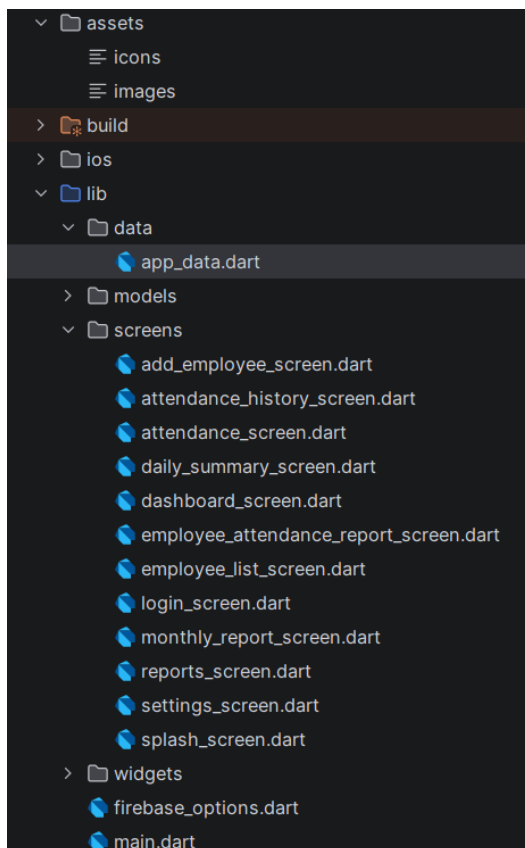
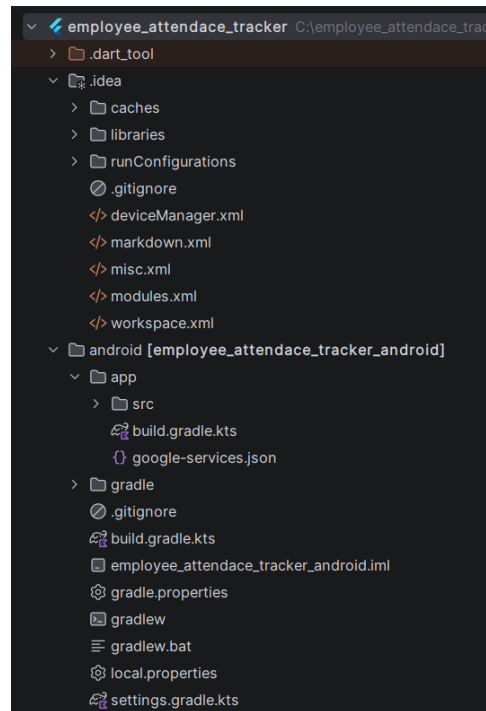
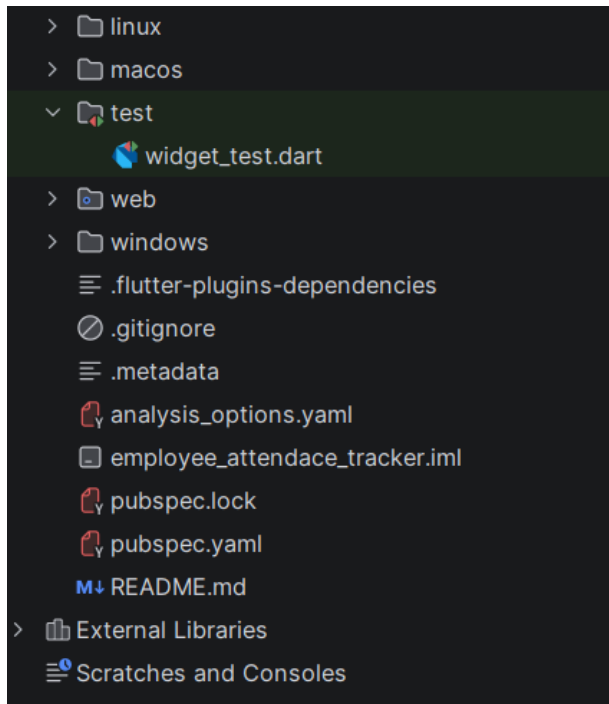
- Add employee
- Update employee
- Delete employee
- Search employee
- View employee details

Technology Stack:

The project is developed using modern mobile application technologies.

Layer	Technology
Frontend	Flutter
Programming Language	Dart
Database	SQLite
IDE	Android Studio
UI Design	Material Design

Folder Structure:



Code:**data/app_data.dart:**

```

import '../models/employee.dart';
import '../models/attendance.dart';
class AppData {
  static final List<Employee> employees = [
    Employee(
      employeeId: 'EMP001',
      name: 'Rahul Sharma',
      department: 'IT',
      designation: 'Developer',
      phone: '9876543210',
      email: 'rahul@gmail.com',
      joiningDate: '01/08/2026',
    ),
    Employee(
      employeeId: 'EMP002',
      name: 'Aman Khan',
      department: 'HR',
      designation: 'Manager',
      phone: '9876543211',
      email: 'aman@gmail.com',
      joiningDate: '05/08/2026',
    ),
    Employee(
      employeeId: 'EMP003',
      name: 'Priya Patel',
      department: 'Finance',
      designation: 'Accountant',
      phone: '9876543212',
      email: 'priya@gmail.com',
      joiningDate: '10/08/2026',
    ),
  ];
  static final List<Attendance> attendanceRecords = [];
}

```

screens/add_employee_screen.dart:

```

import 'package:flutter/material.dart';
import '../data/app_data.dart';
import '../models/employee.dart';

class AddEmployeeScreen extends StatefulWidget {
  final Employee? employee;
  const AddEmployeeScreen({
    super.key,
    this.employee,
  });
  @override
  State<AddEmployeeScreen> createState() =>
    _AddEmployeeScreenState();
}

class _AddEmployeeScreenState extends State<AddEmployeeScreen> {
  final formKey = GlobalKey<FormState>();
  late TextEditingController nameController;
  late TextEditingController departmentController;
  late TextEditingController designationController;
  late TextEditingController phoneController;
  late TextEditingController emailController;
  late TextEditingController joiningDateController;
  bool get isEditing => widget.employee != null;
  @override
  void initState() {
    super.initState();
    final employee = widget.employee;
    nameController = TextEditingController(
      text: employee?.name ?? "",
    );
    departmentController = TextEditingController(
      text: employee?.department ?? "",
    );
    designationController = TextEditingController(
      text: employee?.designation ?? "",
    );
    phoneController = TextEditingController(
      text: employee?.phone ?? "",
    );
    emailController = TextEditingController(
      text: employee?.email ?? "",
    );
  }
}

```

```

joiningDateController = TextEditingController(
    text: employee?.joiningDate ?? "",
);
}
@override
void dispose() {
    nameController.dispose();
    departmentController.dispose();
    designationController.dispose();
    phoneController.dispose();
    emailController.dispose();
    joiningDateController.dispose();
    super.dispose();
}
Future<void> selectJoiningDate() async {
    DateTime initialDate = DateTime.now();
    if (joiningDateController.text.isNotEmpty) {
        final parts = joiningDateController.text.split('/');
        if (parts.length == 3) {
            final day = int.tryParse(parts[0]);
            final month = int.tryParse(parts[1]);
            final year = int.tryParse(parts[2]);
            if (day != null &&
                month != null &&
                year != null) {
                final existingDate =
                    DateTime(year, month, day);
                if (!existingDate.isAfter(DateTime.now())) {
                    initialDate = existingDate;
                }
            }
        }
    }
    final DateTime? pickedDate =
        await showDatePicker(
            context: context,
            initialDate: initialDate,
            firstDate: DateTime(2000),
            lastDate: DateTime.now(),
        );
    if (pickedDate != null) {
        setState() {
            joiningDateController.text =

```

```

        '${pickedDate.day}/'
        '${pickedDate.month}/'
        '${pickedDate.year}';
    });
}
}
void saveEmployee() {
    if (!formKey.currentState!.validate()) {
        return;
    }
    if (isEditing) {
        final employee = widget.employee!;
        employee.name =
            nameController.text.trim();
        employee.department =
            departmentController.text.trim();
        employee.designation =
            designationController.text.trim();
        employee.phone =
            phoneController.text.trim();
        employee.email =
            emailController.text.trim();
        employee.joiningDate =
            joiningDateController.text.trim();
        ScaffoldMessenger.of(context).showSnackBar(
            const SnackBar(
                content: Text(
                    'Employee updated successfully',
                ),
            ),
        );
    }
    else {
        final newId =
            'EMP${(AppData.employees.length + 1).toString().padLeft(3, '0')}';
        final employee = Employee(
            employeeId: newId
            name:
                nameController.text.trim(),
            department:
                departmentController.text.trim(),
            designation:
                designationController.text.trim(),

```

```

    phone:
    phoneController.text.trim(),
    email:
    emailController.text.trim(),
    joiningDate:
    joiningDateController.text.trim(),
  );
  AppData.employees.add(employee);
  ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
      content: Text(
        'Employee added successfully',
      ),
    ),
  );
}
Navigator.pop(context);
}
Widget buildField({
  required TextEditingController controller,
  required String label,
  required IconData icon,
  TextInputType keyboardType =
    TextInputType.text,
}) {
  return Padding(
    padding: const EdgeInsets.only(
      bottom: 15,
    ),
    child: TextFormField(
      controller: controller,
      keyboardType: keyboardType,
      decoration: InputDecoration(
        labelText: label,
        prefixIcon: Icon(icon),
        border:
          const OutlineInputBorder(),
      ),
      validator: (value) {
        final text =
          value?.trim() ?? "";
        if (text.isEmpty) {
          return '$label is required';
        }
      },
    ),
  );
}

```

```

    }
    if (label == 'Phone') {
      if (!RegExp(
        r'^[0-9]{10}$',
      ).hasMatch(text)) {
        return 'Enter a valid 10-digit phone number';
      }
    }
    if (label == 'Email') {
      if (!RegExp(
        r'^[^\s]+@[^\s]+\.[^\s]+$',
      ).hasMatch(text)) {
        return 'Enter a valid email address';
      }
    }
    return null;
  },
),
);
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text(
        isEditing
          ? 'Edit Employee'
          : 'Add Employee',
      ),
      centerTitle: true,
    ),
    body: SingleChildScrollView(
      padding:
        const EdgeInsets.all(16),
      child: Form(
        key: formKey,
        child: Column(
          children: [
            buildField(
              controller:
                nameController,
              label:
                'Employee Name',

```

```

        icon:
        Icons.person,
    ),
    buildField(
        controller:
        departmentController,
        label:
        'Department',
        icon:
        Icons.business,
    ),
    buildField(
        controller:
        designationController,
        label:
        'Designation',
        icon:
        Icons.work,
    ),
    buildField(
        controller:
        phoneController,
        label:
        'Phone',
        icon:
        Icons.phone,
        keyboardType:
        TextInputType.phone,
    ),
    buildField(
        controller:
        emailController,
        label:
        'Email',
        icon:
        Icons.email,
        keyboardType:
        TextInputType.emailAddress,
    ),
    Padding(
        padding:
        const EdgeInsets.only(
            bottom: 15,

```

```

),
child: TextFormField(
  controller:
    joiningDateController,
  readOnly: true,
  onTap:
    selectJoiningDate,
  decoration:
    const InputDecoration(
      labelText:
        'Joining Date',
      hintText:
        'Select joining date',
      prefixIcon:
        Icon(
          Icons.calendar_today,
        ),
      suffixIcon:
        Icon(
          Icons.arrow_drop_down,
        )
      border:
        OutlineInputBorder(),
    ),
  validator: (value) {
    if (value == null ||
      value.trim().isEmpty) {
      return 'Joining Date is required';
    }
    return null;
  },
),
),
const SizedBox(
  height: 10,
),
SizedBox(
  width:
    double.infinity,
  height: 55,
  child:
    ElevatedButton.icon(
      onPressed:

```

```

        saveEmployee,
        icon: Icon(
          isEditing
            ? Icons.update
            : Icons.save,
        ),
        label: Text(
          isEditing
            ? 'Update Employee'
            : 'Save Employee',
        ),
      ),
    ],
  ),
),
);

```

screens/attendance_history_screen.dart:

```

import 'package:flutter/material.dart';
import '../data/app_data.dart';
import '../models/attendance.dart';
class AttendanceHistoryScreen extends StatefulWidget {
  const AttendanceHistoryScreen({super.key});
  @override
  State<AttendanceHistoryScreen> createState() =>
    _AttendanceHistoryScreenState();
}
class _AttendanceHistoryScreenState
  extends State<AttendanceHistoryScreen> {
  void editAttendance(Attendance record) {
    String selectedStatus = record.status;
    showDialog(
      context: context,
      builder: (dialogContext) {
        return StatefulBuilder(
          builder: (
            context,

```

```

    setDialogState,
  ) {
return AlertDialog(
  title: const Text(
    'Edit Attendance',
  ),
  content: Column(
    mainAxisAlignment: MainAxisAlignment.min,
    crossAxisAlignment:
    CrossAxisAlignment.start,
    children: [
      const Text(
        'Employee',
        style: TextStyle(
          fontWeight:
          FontWeight.bold,
        ),
      ),
      const SizedBox(height: 5),
      Text(
        '${record.employeeId} - '
        '${record.employeeName}',
      ),
      const SizedBox(height: 15),
      const Text(
        'Date',
        style: TextStyle(
          fontWeight:
          FontWeight.bold,
        ),
      ),
      const SizedBox(height: 5),
      Text(
        record.date,
      ),
      const SizedBox(height: 20),
      const Text(
        'Attendance Status',
        style: TextStyle(
          fontWeight:
          FontWeight.bold,
        ),
      ),
    ],
  ),
);

```

```

RadioListTile<String>(
  contentPadding:
    EdgeInsets.zero,
  title:
    const Text('Present'),
  value: 'Present',
  groupValue:
    selectedStatus,
  onChanged: (value) {
    setDialogState() {
      selectedStatus =
        value!;
    });
  });
),
RadioListTile<String>(
  contentPadding:
    EdgeInsets.zero,
  title:
    const Text('Absent'),
  value: 'Absent',
  groupValue:
    selectedStatus
  onChanged: (value) {
    setDialogState() {
      selectedStatus =
        value!;
    });
  });
),
RadioListTile<String>(
  contentPadding:
    EdgeInsets.zero,
  title:
    const Text('Leave'),
  value: 'Leave',
  groupValue:
    selectedStatus,
  onChanged: (value) {
    setDialogState() {
      selectedStatus =
        value!;
    });
  });

```

```

    },
  ),
],
),
actions: [
  TextButton(
    onPressed: () {
      Navigator.pop(
        dialogContext, ); },
    child:
      const Text('Cancel'),
  ),
  ElevatedButton(
    onPressed: () {
      setState(() {
        record.status =
          selectedStatus;
      });
      Navigator.pop(
        dialogContext,
      );
      ScaffoldMessenger.of(
        context,
      ).showSnackBar(
        const SnackBar(
          content: Text(
            'Attendance updated successfully',
          ),
        ),
      );
    },
    child:
      const Text('Update'),
  ),
],
);
},
);
},
);
}
void deleteAttendance(
  Attendance record) {

```

```

showDialog(
  context: context,
  builder: (context) {
    return AlertDialog(
      title: const Text(
        'Delete Attendance',
      ),
      content: Text(
        'Delete attendance record for '
        '${record.employeeName} '
        'on ${record.date}?',
      ),
      actions: [
        TextButton(
          onPressed: () {
            Navigator.pop(context);
          },
          child:
            const Text('Cancel'),
        ),
        TextButton(
          onPressed: () {
            setState(() {
              AppData.attendanceRecords
                .remove(record);
            });
            Navigator.pop(context);
            ScaffoldMessenger.of(
              this.context,
            ).showSnackBar(
              const SnackBar(
                content: Text(
                  'Attendance deleted successfully',
                ),
              ),
            );
          },
          child: const Text(
            'Delete',
            style: TextStyle(
              color: Colors.red,
            ),
          ),
        ),
      ],
    );
  },
);

```

```

    ),
  ],
);
},
);
}
@override
Widget build(BuildContext context) {
  final records =
    AppData.attendanceRecords;
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'Attendance History',
      ),
      centerTitle: true,
    ),
    body: records.isEmpty
      ? const Center(
        child: Column(
          mainAxisAlignment:
            MainAxisAlignment.center,
          children: [
            Icon(
              Icons.history,
              size: 60,
              color: Colors.grey,
            ),
            SizedBox(height: 12)
            Text(
              'No attendance records found',
              style: TextStyle(
                fontSize: 18,
                color: Colors.grey,
              ),
            ),
          ],
        ),
      )
      : ListView.builder(
        padding:
          const EdgeInsets.all(16)
        itemCount:
          records.length,

```

```

itemBuilder:
  (context, index) {
    final record =
      records[index];
    return Card(
      elevation: 3,
      margin:
        const EdgeInsets.only(
          bottom: 12,
        ),
      child: ListTile(
        leading: CircleAvatar(
          child: Text(
            record.employeeName[0]
              .toUpperCase(),
          ),
        ),
        title: Text(
          record.employeeName,
          style:
            const TextStyle(
              fontWeight:
                FontWeight.bold,
            ),
        ),
        subtitle: Text(
          '${record.employeeId}\n'
            '${record.date}',
        ),
        isThreeLine: true,
        trailing: Row(
          mainAxisAlignment:
            MainAxisAlignment.min,
          children: [
            Column
              mainAxisAlignment:
                MainAxisAlignment.center,
              children: [
                Text(
                  record.status,
                  style: TextStyle(
                    fontWeight:
                      FontWeight.bold,

```

```

        color:
        record.status ==
            'Present'
            ? Colors.green
            : record.status ==
            'Absent'
            ? Colors.red
            : Colors.orange,
    ),
),
],
),
IconButton(
  onPressed: () {
    editAttendance(
      record,
    );
  },
  icon:
  const Icon(
    Icons.edit,
  ),
  tooltip:
  'Edit Attendance',
),
IconButton(
  onPressed: () {
    deleteAttendance(
      record,
    );
  },
  icon:
  const Icon(
    Icons.delete,
  ),
  tooltip:
  'Delete Attendance',
),
],
),
),
);

```

screens/attendance_screen.dart:

```

import 'package:flutter/material.dart';
import '../data/app_data.dart';
import '../models/attendance.dart';
import '../models/employee.dart';
class AttendanceScreen extends StatefulWidget {
  const AttendanceScreen({super.key});
  @override
  State<AttendanceScreen> createState() =>
    _AttendanceScreenState();
}
class _AttendanceScreenState
  extends State<AttendanceScreen> {
  Employee? selectedEmployee;
  String selectedStatus = 'Present';
  DateTime selectedDate = DateTime.now();
  String get formattedDate {
    return '${selectedDate.day}/'
      '${selectedDate.month}/'
      '${selectedDate.year}';
  }
  Future<void> selectDate() async {
    final DateTime? pickedDate =
    await showDatePicker(
      context: context,
      initialDate: selectedDate,
      firstDate: DateTime(2020),
      lastDate: DateTime.now(),
    );
    if (pickedDate != null) {
      setState() {
        selectedDate = pickedDate;
      });
    }
  }
  void saveAttendance() {
    if (selectedEmployee == null) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(
          content: Text(
            'Please select an employee',

```

```

    ),
  ),
);
return;
}
final alreadyMarked =
AppData.attendanceRecords.any(
  (record) =>
    record.employeeId ==
      selectedEmployee!.employeeId &&
      record.date == formattedDate,
);
if (alreadyMarked) {
  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(
      content: Text(
        'Attendance already marked for '
        '${selectedEmployee!.name} '
        'on $formattedDate',
      ),
    ),
  );
return;
}
final attendance = Attendance(
  employeeId:
    selectedEmployee!.employeeId,
  employeeName:
    selectedEmployee!.name,
  date: formattedDate,
  status: selectedStatus,
);
setState() {
  AppData.attendanceRecords.add(
    attendance,
  );
});
ScaffoldMessenger.of(context).showSnackBar(
  const SnackBar(
    content: Text(
      'Attendance saved successfully',
    ),
  ),
);

```

```

);
setState(() {
  selectedEmployee = null;
  selectedStatus = 'Present';
});
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'Mark Attendance',
      ),
      centerTitle: true,
    ),
    body: SingleChildScrollView(
      padding:
        const EdgeInsets.all(16),
      child: Column(
        crossAxisAlignment:
          CrossAxisAlignment.start,
        children: [
          const Text(
            'Mark Employee Attendance',
            style: TextStyle(
              fontSize: 24,
              fontWeight:
                FontWeight.bold,
            ),
          ),
          const SizedBox(height: 5),
          const Text(
            'Select employee, date and attendance status.',
            style: TextStyle(
              color: Colors.grey,
            ),
          ),
          const SizedBox(height: 25),
          const Text(
            'Select Employee',
            style: TextStyle(
              fontSize: 18,
              fontWeight:

```

```

    FontWeight.bold,
  ),
),
const SizedBox(height: 10),
DropDownButtonFormField<Employee>(
  value: selectedEmployee
  decoration:
  const InputDecoration(
    border:
    OutlineInputBorder(),
    prefixIcon:
    Icon(Icons.person),
    labelText:
    'Employee',
  ),
  hint: const Text(
    'Choose employee',
  ),
  items:
  AppData.employees
    .map(
      (employee) {
        return DropdownMenuItem<
          Employee>(
            value: employee,
            child: Text(
              '${employee.employeeId} - '
              '${employee.name}',
            ),
          ),
      );
    },
  ).toList(),
  onChanged:
    (employee) {
      setState(() {
        selectedEmployee =
          employee;
      });
    },
),
const SizedBox(height: 25),
const Text(
  'Attendance Date',
  style: TextStyle(

```

```

        fontSize: 18,
        fontWeight:
        FontWeight.bold,
    ),
),
const SizedBox(height: 10),
InkWell(
  onTap: selectDate
  child: InputDecorator(
    decoration:
    const InputDecoration(
      border:
      OutlineInputBorder(),
      prefixIcon:
      Icon(
        Icons.calendar_today,
      ),
      labelText:
      'Date',
    ),
    child: Text(
      formattedDate,
      style:
      const TextStyle(
        fontSize: 16,
      ),
    ),
  ),
),
const SizedBox(height: 25),
const Text(
  'Attendance Status',
  style: TextStyle(
    fontSize: 18,
    fontWeight:
    FontWeight.bold,
  ),
),
const SizedBox(height: 10),
Card(
  child: RadioListTile<String>(
    title: const Text(
      'Present',

```

```

    ),
    secondary:
    const Icon(
      Icons.check_circle,
      color: Colors.green,
    ),
    value: 'Present',
    groupValue:
    selectedStatus,
    onChanged:
      (value) {
        setState(() {
          selectedStatus =
            value!;
        });
      },
    ),
  ),
  Card(
    child: RadioListTile<String>(
      title: const Text(
        'Absent',
      ),
      secondary:
      const Icon(
        Icons.cancel,
        color: Colors.red,
      ),
      value: 'Absent',
      groupValue:
      selectedStatus,
      onChanged:
        (value) {
          setState(() {
            selectedStatus =
              value!;
          });
        },
    ),
  ),
  Card(
    child: RadioListTile<String>(
      title: const Text(

```

```

      'Leave',
    ),
    secondary:
    const Icon(
      Icons.event_busy,
      color: Colors.orange,
    ),
    value: 'Leave',
    groupValue:
    selectedStatus,
    onChanged:
      (value) {
        setState(() {
          selectedStatus =
            value!;
        });
      },
    ),
  ),
  const SizedBox(height: 25),
  SizedBox(
    width:
    double.infinity,
    height: 55,
    child:
    ElevatedButton.icon(
      onPressed:
        saveAttendance,
      icon: const Icon(
        Icons.save,
      ),
      label: const Text(
        'Save Attendance',
        style: TextStyle(
          fontSize: 16,
        ),
      ),
    ),
  ),
)
const SizedBox(height: 15),
Container(
  width:
  double.infinity,

```

```

padding:
const EdgeInsets.all(12),
decoration:
BoxDecoration(
  borderRadius:
    BorderRadius.circular(
      8,
    ),
  color:
    Colors.blue.withValues(
      alpha: 0.08,
    ),
),
child: const Row(
  crossAxisAlignment:
    CrossAxisAlignment.start,
  children: [
    Icon(
      Icons.info_outline,
      color: Colors.blue,
    ),
    SizedBox(width: 10),
    Expanded(
      child: Text(
        'Only one attendance record '
        'can be added for an employee '
        'on the same date.',
      ),
    ),
  ],
),
),
],
),
),
);
}
}

```

screens/daily_summary_screen.dart:

```

import 'package:flutter/material.dart';
import '../data/app_data.dart';
class DailySummaryScreen extends StatefulWidget {
  const DailySummaryScreen({super.key});
  @override
  State<DailySummaryScreen> createState() =>
    _DailySummaryScreenState();
}
class _DailySummaryScreenState extends State<DailySummaryScreen> {
  DateTime selectedDate = DateTime.now();
  String get formattedDate {
    return '${selectedDate.day}/${selectedDate.month}/${selectedDate.year}';
  }
  Future<void> selectDate() async {
    final DateTime? pickedDate = await showDatePicker(
      context: context,
      initialDate: selectedDate,
      firstDate: DateTime(2020),
      lastDate: DateTime.now(),
    );
    if (pickedDate != null) {
      setState(() {
        selectedDate = pickedDate;
      });
    }
  }
  @override
  Widget build(BuildContext context) {
    final records = AppData.attendanceRecords
      .where(
        (record) => record.date == formattedDate,
      )
      .toList();
    final present = records
      .where(
        (record) => record.status == 'Present',
      )
      .length;
    final absent = records
      .where(

```

```

      (record) => record.status == 'Absent',
    )
    .length
  final leave = records
    .where(
      (record) => record.status == 'Leave',
    )
    .length;
  return Scaffold(
    appBar: AppBar(
      title: const Text('Daily Summary'),
      centerTitle: true,
    ),
    body: Padding(
      padding: const EdgeInsets.all(16),
      child: Column(
        children: [
          InkWell(
            onTap: selectDate,
            borderRadius: BorderRadius.circular(12),
            child: Card(
              child: Padding(
                padding: const EdgeInsets.all(20),
                child: Row(
                  children: [
                    const Icon(
                      Icons.calendar_month,
                      size: 40,
                    ),
                    const SizedBox(width: 20),
                    Column(
                      crossAxisAlignment:
                        CrossAxisAlignment.start,
                      children: [
                        const Text(
                          'Selected Date',
                          style: TextStyle(
                            fontSize: 15,
                            color: Colors.grey,
                          ),
                        ),
                        const SizedBox(height: 5),
                        Text(

```

```

        formattedDate,
        style: const TextStyle(
          fontSize: 22,
          fontWeight: FontWeight.bold,
        ),
      ),
    ],
  ),
  const Spacer(),
  const Icon(
    Icons.arrow_drop_down,
    size: 30,
  ),
],
),
),
),
),
const SizedBox(height: 20),
Row(
  children: [
    Expanded(
      child: _summaryCard(
        'Present',
        present.toString(),
        Icons.check_circle,
        Colors.green,
      ),
    ),
    const SizedBox(width: 10),
    Expanded(
      child: _summaryCard(
        'Absent',
        absent.toString(),
        Icons.cancel,
        Colors.red,
      ),
    ),
    const SizedBox(width: 10),
    Expanded(
      child: _summaryCard(
        'Leave',
        leave.toString(),

```

```

        Icons.event_busy,
        Colors.orange,
      ),
    ),
  ],
),
const SizedBox(height: 25),
Align(
  alignment: Alignment.centerLeft,
  child: Text(
    'Attendance Records',
    style: const TextStyle(
      fontSize: 20,
      fontWeight: FontWeight.bold,
    ),
  ),
),
const SizedBox(height: 10),
Expanded(
  child: records.isEmpty
    ? Center(
        child: Text(
          'No attendance records found\n'
            'for $formattedDate',
          textAlign: TextAlign.center,
          style: const TextStyle(
            fontSize: 17,
            color: Colors.grey,
          ),
        ),
      ),
    : ListView.builder(
        itemCount: records.length,
        itemBuilder: (context, index) {
          final record = records[index];
          return Card(
            margin: const EdgeInsets.only(
              bottom: 10,
            ),
            child: ListTile(
              leading: CircleAvatar(
                child: Text(
                  record.employeeName[0]

```

```

        .toUpperCase(),
      ),
    ),
    title: Text(
      record.employeeName,
      style: const TextStyle(
        fontWeight: FontWeight.bold,
      ),
    ),
    subtitle: Text(
      record.employeeId,
    ),
    trailing: Text(
      record.status,
      style: TextStyle(
        fontWeight: FontWeight.bold,
        color:
          record.status == 'Present'
            ? Colors.green
            : record.status == 'Absent'
            ? Colors.red
            : Colors.orange,
      ),
    ),
  ),
);
},
),
),
],
),
),
);
}
Widget _summaryCard(
  String title,
  String value,
  IconData icon,
  Color color,
) {
  return Card(
    child: Padding(
      padding: const EdgeInsets.all(12),

```

```

child: Column(
  children: [
    Icon(
      icon,
      color: color,
      size: 28,
    ),
    const SizedBox(height: 6),
    Text(
      value,
      style: const TextStyle(
        fontSize: 22,
        fontWeight: FontWeight.bold,
      ),
    ),
    Text(title),
  ],
)

```

screens/dashboard_screen.dart:

```

import 'package:flutter/material.dart';
import '../data/app_data.dart';
import 'employee_list_screen.dart';
import 'attendance_screen.dart';
import 'attendance_history_screen.dart';
import 'daily_summary_screen.dart';
import 'monthly_report_screen.dart';
import 'employee_attendance_report_screen.dart';
class DashboardScreen extends StatefulWidget {
  const DashboardScreen({super.key});
  @override
  State<DashboardScreen> createState() =>
    _DashboardScreenState();
}
class _DashboardScreenState extends State<DashboardScreen> {
  String getTodayDate() {
    final now = DateTime.now();
    return '${now.day}/${now.month}/${now.year}';
  }
  int get employeeCount {
    return AppData.employees.length;
  }
}

```

```

int get presentCount {
  final today = getTodayDate();
  return AppData.attendanceRecords
    .where(
      (record) =>
        record.date == today &&
        record.status == 'Present',
    )
    .length;
}
int get absentCount {
  final today = getTodayDate();
  return AppData.attendanceRecords
    .where(
      (record) =>
        record.date == today &&
        record.status == 'Absent',
    )
    .length;
}
int get leaveCount {
  final today = getTodayDate();
  return AppData.attendanceRecords
    .where(
      (record) =>
        record.date == today &&
        record.status == 'Leave',
    )
    .length;
}
int get attendancePercentage {
  if (employeeCount == 0) {
    return 0;
  }
  return ((presentCount / employeeCount) * 100)
    .round();
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'Attendance Dashboard',

```

```

),
centerTitle: true,
),
body: RefreshIndicator(
  onRefresh: () async {
    setState(() {});
  },
child: SingleChildScrollView(
  physics:
    const AlwaysScrollableScrollPhysics(),
  padding:
    const EdgeInsets.all(16),
  child: Column(
    crossAxisAlignment:
      CrossAxisAlignment.start,
    children: [
      const Text(
        'Welcome, Admin',
        style: TextStyle(
          fontSize: 24,
          fontWeight:
            FontWeight.bold,
        ),
      ),
      const SizedBox(height: 5),
      const Text(
        'Manage your employees and attendance',
        style: TextStyle(
          color: Colors.grey,
        ),
      ),
      const SizedBox(height: 8),
      Text(
        'Today: ${getTodayDate()}',
        style: const TextStyle(
          fontSize: 14,
          color: Colors.grey,
        ),
      ),
      const SizedBox(height: 25),
      const Text(
        'Today\'s Summary',
        style: TextStyle(

```

```

        fontSize: 20,
        fontWeight:
        FontWeight.bold,
    ),
),
const SizedBox(height: 15),
Row(
  children: [
    Expanded(
      child: _summaryCard(
        'Employees',
        employeeCount.toString(),
        Icons.people,
        Colors.blue,
      ),
    ),
    const SizedBox(width: 12),
    Expanded(
      child: _summaryCard(
        'Present',
        presentCount.toString(),
        Icons.check_circle,
        Colors.green,
      ),
    ),
  ],
),
const SizedBox(height: 12),
Row(
  children: [
    Expanded(
      child: _summaryCard(
        'Absent',
        absentCount.toString(),
        Icons.cancel,
        Colors.red,
      ),
    ),
    const SizedBox(width: 12),
    Expanded(
      child: _summaryCard(
        'Leave',
        leaveCount.toString(),
        Icons.event_busy,
        Colors.orange,

```

```

    ),
  ),
],
),
const SizedBox(height: 12),
  _attendancePercentageCard(),
const SizedBox(height: 30),
const Text(
  'Quick Actions',
  style: TextStyle(
    fontSize: 20,
    fontWeight:
      FontWeight.bold,
  ),
),
const SizedBox(height: 15),
  _actionButton(
    context,
    title: 'Manage Employees',
    icon: Icons.people,
    screen:
      const EmployeeListScreen(),
  ),
const SizedBox(height: 12),
  _actionButton(
    context,
    title: 'Mark Attendance',
    icon: Icons.check_circle,
    screen:
      const AttendanceScreen(),
  ),
const SizedBox(height: 12)
  _actionButton(
    context,
    title: 'Attendance History',
    icon: Icons.history,
    screen:
      const AttendanceHistoryScreen(),
  ),
const SizedBox(height: 12),
  _actionButton(
    context,
    title: 'Daily Summary',

```

```

        icon: Icons.bar_chart,
        screen:
        const DailySummaryScreen(),
    ),
    const SizedBox(height: 12),
    _actionButton(
        context,
        title: 'Monthly Report',
        icon: Icons.calendar_month,
        screen:
        const MonthlyReportScreen(),
    ),
    const SizedBox(height: 12),
    _actionButton(
        context,
        title: 'Employee Reports',
        icon: Icons.person_search,
        screen:
        const EmployeeAttendanceReportScreen(),
    ),
    const SizedBox(height: 20),
    Center(
        child: Text(
            'Employee Attendance Tracker',
            style: TextStyle(
                color: Colors.grey.shade600,
                fontSize: 13,
            ),
        ),
    ),
    const SizedBox(height: 10),
],
),
),
),
);
}
Widget _summaryCard(
    String title,
    String value,
    IconData icon,
    Color color,
) {

```

```

return Card(
  elevation: 3,
  child: Padding(
    padding:
      const EdgeInsets.all(16),
    child: Column(
      children: [
        Icon(
          icon,
          size: 32,
          color: color,
        ),
        const SizedBox(height: 8),
        Text(
          value,
          style: const TextStyle(
            fontSize: 22,
            fontWeight:
              FontWeight.bold,
          ),
        ),
        const SizedBox(height: 5),
        Text(
          title,
          textAlign:
            TextAlign.center,
        ),
      ],
    ),
  ),
);
}

Widget _attendancePercentageCard() {
  return Card(
    elevation: 3,
    child: Padding(
      padding:
        const EdgeInsets.all(16),
      child: Row(
        children: [
          CircleAvatar(
            radius: 28,
            child: Text(

```

```

        '$attendancePercentage%',
        style: const TextStyle(
          fontWeight:
            FontWeight.bold,
        ),
      ),
    ),
    const SizedBox(width: 15),
    const Expanded(
      child: Column(
        crossAxisAlignment:
          CrossAxisAlignment.start,
        children: [
          Text(
            'Today\'s Attendance',
            style: TextStyle(
              fontSize: 17,
              fontWeight:
                FontWeight.bold,
            ),
          ),
          SizedBox(height: 5),
          Text(
            'Percentage of employees marked Present today.',
            style: TextStyle(
              color: Colors.grey,
            ),
          ),
        ],
      ),
    ),
  ],
),
);
}
Widget _actionButton(
  BuildContext context,
  required String title,
  required IconData icon
  required Widget screen,
) {
  return SizedBox(

```

```

width:
double.infinity,
height: 55,
child: ElevatedButton.icon(
  onPressed: () async {
    await Navigator.push(
      context,
      MaterialPageRoute(
        builder: (context) =>
          screen,
      ),
    );
    setState(() {});
  },
  icon: Icon(icon),
  label: Text(
    title,
    style: const TextStyle(
      fontSize: 15,
    ),
  ),
),
);
}
}

```

screens/employee_attendance_report_screen.dart:

```

import 'package:flutter/material.dart';
import '../data/app_data.dart';
class EmployeeAttendanceReportScreen extends StatefulWidget {
  const EmployeeAttendanceReportScreen({super.key});

  @override
  State<EmployeeAttendanceReportScreen> createState() =>
    _EmployeeAttendanceReportScreenState();
}
class _EmployeeAttendanceReportScreenState
  extends State<EmployeeAttendanceReportScreen> {
  final TextEditingController searchController =
    TextEditingController();
  String selectedFilter = 'All';
  @override
  void dispose() {
    searchController.dispose();
    super.dispose();
  }

  List<dynamic> get filteredEmployees {
    final searchText =
      searchController.text.toLowerCase().trim();
    return AppData.employees.where((employee) {
      final matchesSearch =
        employee.name.toLowerCase().contains(searchText) ||
        employee.employeeId
          .toLowerCase()
          .contains(searchText) ||
        employee.department
          .toLowerCase()
          .contains(searchText);
      if (!matchesSearch) {
        return false;
      }
      if (selectedFilter == 'All') {
        return true;
      }
    });
    final hasStatus = AppData.attendanceRecords.any(

```

```

        (record) =>
        record.employeeId == employee.employeeId &&
        record.status == selectedFilter,
    );
    return hasStatus;
  }).toList();
}
@override
Widget build(BuildContext context) {
  final employees = filteredEmployees;
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'Employee Attendance Report',
      ),
      centerTitle: true,
    ),
    body: Column(
      children: [
        Padding(
          padding: const EdgeInsets.fromLTRB(
            16,
            16,
            16,
            8,
          ),
          child: TextField(
            controller: searchController,
            onChanged: (value) {
              setState(() {});
            },
            decoration: InputDecoration(
              hintText:
                'Search employee, ID or department',
              prefixIcon: const Icon(
                Icons.search,
              ),
              suffixIcon: searchController.text.isEmpty
                ? null
                : IconButton(
                  onPressed: () {
                    searchController.clear();
                    setState(() {});

```

```

    },
    icon: const Icon(
      Icons.clear,
    ),
  ),
  border: const OutlineInputBorder(),
  filled: true,
),
),
),

Padding(
  padding: const EdgeInsets.symmetric(
    horizontal: 16,
    vertical: 8,
  ),
  child: DropdownButtonFormField<String>(
    value: selectedFilter,
    decoration: const InputDecoration(
      labelText: 'Filter by Attendance Status',
      border: OutlineInputBorder(),
      prefixIcon: Icon(
        Icons.filter_list,
      ),
    ),
    items: const [
      DropdownMenuItem(
        value: 'All',
        child: Text('All Employees'),
      ),
      DropdownMenuItem(
        value: 'Present',
        child: Text('Present'),
      ),
      DropdownMenuItem(
        value: 'Absent',
        child: Text('Absent'),
      ),
      DropdownMenuItem(
        value: 'Leave',
        child: Text('Leave'),
      ),
    ],
  ),
),

```

```

onChanged: (value) {
  setState(() {
    selectedFilter = value!;
  });
},
),
),
const SizedBox(height: 8),
Padding(
  padding: const EdgeInsets.symmetric(
    horizontal: 16,
  ),
  child: Align(
    alignment: Alignment.centerLeft,
    child: Text(
      '${employees.length} employee(s) found',
      style: const TextStyle(
        fontWeight: FontWeight.bold,
        color: Colors.grey,
      ),
    ),
  ),
),
),
const SizedBox(height: 8),
Expanded(
  child: employees.isEmpty
    ? const Center(
      child: Column(
        mainAxisAlignment:
          MainAxisAlignment.center,
        children:
          Icon(
            Icons.search_off,
            size: 60,
            color: Colors.grey,
          ),
        SizedBox(height: 12),
        Text(
          'No employees found',
          style: TextStyle(
            fontSize: 18,
            color: Colors.grey,
          ),
        ),
      ),
    ),
  ),
),

```

```

        ),
      ],
    ),
  )
  : ListView.builder(
    padding: const EdgeInsets.all(16),
    itemCount: employees.length,
    itemBuilder: (context, index) {
      final employee =
        employees[index];
      return _employeeCard(
        employee,
      );
    },
  ),
),
],
),
);
}
Widget _employeeCard(dynamic employee) {
  final records = AppData.attendanceRecords
    .where(
      (record) =>
        record.employeeId ==
          employee.employeeId,
    )
    .toList();
  final present = records
    .where(
      (record) =>
        record.status == 'Present',
    )
    .length;
  final absent = records
    .where(
      (record) =>
        record.status == 'Absent',
    )
    .length;
  final leave = records
    .where(
      (record) =>

```

```

    record.status == 'Leave',
  )
    .length;
final total = records.length;
final percentage = total == 0
  ? 0
  : ((present / total) * 100).round();
return Card(
  elevation: 3,
  margin: const EdgeInsets.only(
    bottom: 15,
  ),
  child: Padding(
    padding: const EdgeInsets.all(16),
    child: Column(
      crossAxisAlignment:
        CrossAxisAlignment.start,
      children: [
        Row(
          children: [
            CircleAvatar(
              radius: 25,
              child: Text(
                employee.name[0]
                  .toUpperCase(),
                style: const TextStyle(
                  fontSize: 20,
                  fontWeight:
                    FontWeight.bold,
                ),
              ),
            ),
            const SizedBox(width: 12),
            Expanded(
              child: Column(
                crossAxisAlignment:
                  CrossAxisAlignment.start
                children: [
                  Text(
                    employee.name,
                    style: const TextStyle(
                      fontSize: 18,
                      fontWeight:

```

```

        FontWeight.bold,
      ),
    ),
    const SizedBox(height: 4)
    Text(
      employee.employeeId
      style: const TextStyle(
        color: Colors.grey,
      ),
    ),
    Text(
      '${employee.department} • '
      '${employee.designation}',
      style: const TextStyle(
        color: Colors.grey,
      ),
    ),
  ],
),
),
],
)
const SizedBox(height: 20),
const Divider(),
const SizedBox(height: 10),

Row(
  children: [
    Expanded(
      child: _statItem(
        'Present',
        present.toString(),
        Colors.green,
      ),
    ),
    Expanded(
      child: _statItem(
        'Absent',
        absent.toString(),
        Colors.red,
      ),
    ),
    Expanded(

```

```

        child: _statItem(
          'Leave',
          leave.toString(),
          Colors.orange,
        ),
      ),
      Expanded(
        child: _statItem(
          'Total',
          total.toString(),
          Colors.blue,
        ),
      ),
    ],
  ),
  const SizedBox(height: 20),
  const Text(
    'Attendance Percentage',
    style: TextStyle(
      fontWeight: FontWeight.bold,
    ),
  )
  const SizedBox(height: 8),
  LinearProgressIndicator(
    value: percentage / 100,
    minHeight: 10,
  )
  const SizedBox(height: 8),
  Align(
    alignment: Alignment.centerRight,
    child: Text(
      '$percentage%',
      style: const TextStyle(
        fontSize: 18,
        fontWeight:
          FontWeight.bold,
      ),
    ),
  ),
],
),
),
);

```

```
}  
Widget _statItem(  
  String title,  
  String value,  
  Color color,  
) {  
  return Column(  
    children: [  
      Text(  
        value,  
        style: TextStyle(  
          fontSize: 22,  
          fontWeight:  
            FontWeight.bold,  
          color: color,  
        ),  
      ),  
      const SizedBox(height: 4),  
      Text(  
        title,  
        style: const TextStyle(  
          fontSize: 12,  
        ),  
      ),  
    ],  
  ),  
}
```

screens/employee_list_screen.dart:

```
import 'package:flutter/material.dart';
import '../data/app_data.dart';
import '../models/employee.dart';
import 'add_employee_screen.dart';

class EmployeeListScreen extends StatefulWidget {
  const EmployeeListScreen({super.key});

  @override
  State<EmployeeListScreen> createState() =>
    _EmployeeListScreenState();
}

class _EmployeeListScreenState
  extends State<EmployeeListScreen> {
  void deleteEmployee(Employee employee) {
    showDialog(
      context: context,
      builder: (context) {
        return AlertDialog(
          title: const Text(
            'Delete Employee',
          ),
          content: Text(
            'Are you sure you want to delete '
              '${employee.name}?',
          ),
          actions: [
            TextButton(
              onPressed: () {
                Navigator.pop(context);
              },
              child: const Text(
                'Cancel',
              ),
            ),
            TextButton(
              onPressed: () {
                setState(() {
                  AppData.employees.remove(employee);
                });
              },
              child: const Text(
                'Delete',
              ),
            ),
          ],
        );
      },
    );
  }
}
```

```

    });
    Navigator.pop(context);
    ScaffoldMessenger.of(context)
      .showSnackBar(
        const SnackBar(
          content: Text(
            'Employee deleted successfully',
          ),
        ),
      );
  },
  child: const Text(
    'Delete',
    style: TextStyle(
      color: Colors.red,
    ),
  ),
),
],
);
},
);
}
void editEmployee(Employee employee) {
  Navigator.push(
    context,
    MaterialPageRoute(
      builder: (context) => AddEmployeeScreen(
        employee: employee,
      ),
    ),
  ).then((value) {
    setState(() {});
  });
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'Employees',
      ),
      centerTitle: true,

```

```

),
body: AppData.employees.isEmpty
  ? const Center(
child: Text(
  'No employees found',
  style: TextStyle(
    fontSize: 18,
    color: Colors.grey,
  ),
),
),
)
: ListView.builder(
padding: const EdgeInsets.all(16),
itemCount:
AppData.employees.length,
itemBuilder:
  (context, index) {
    final employee =
AppData.employees[index];
    return Card(
      elevation: 3,
      margin:
const EdgeInsets.only(
        bottom: 12,
      ),
      child: ListTile(
        contentPadding:
const EdgeInsets.all(12),
        leading: CircleAvatar(
          radius: 25,
          child: Text(
            employee.name[0]
              .toUpperCase(),
            style: const TextStyle(
              fontSize: 20,
              fontWeight:
                FontWeight.bold,
            ),
          ),
        ),
        title: Text
          employee.name,
        style:

```

```

const TextStyle(
  fontSize: 17,
  fontWeight:
    FontWeight.bold,
),
),
subtitle: Column(
  crossAxisAlignment:
    CrossAxisAlignment.start,
  children: [
    const SizedBox(
      height: 4,
    ),
    Text(
      employee.employeeId,
    ),
    Text(
      '${employee.department} • '
      '${employee.designation}',
    ),
    Text(
      employee.phone,
    ),
  ],
),
trailing: Row(
  mainAxisAlignment:
    MainAxisAlignment.min,
  children: [
    IconButton(
      onPressed: () {
        editEmployee(
          employee,
        );
      },
      icon: const Icon(
        Icons.edit,
      ),
      tooltip:
        'Edit Employee',
    ),
    IconButton(
      onPressed: () {

```

```

        deleteEmployee(
          employee,
        );
      },
      icon: const Icon(
        Icons.delete,
      ),
      tooltip:
        'Delete Employee',
    ),
  ],
),
),
);
},
),
floatingActionButton:
FloatingActionButton.extended
onPressed: () {
  Navigator.push(
    context
    MaterialPageRoute(
      builder: (context) =>
        const AddEmployeeScreen(),
    ),
  ).then((value) {
    setState(() {});
  });
},
icon: const Icon(
  Icons.person_add,
),
label: const Text(
  'Add Employee', ),

```

screens/login_screen.dart:

```

import 'package:flutter/material.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'dashboard_screen.dart';
class LoginScreen extends StatefulWidget {
  const LoginScreen({super.key});
  @override
  State<LoginScreen> createState() {
    return _LoginScreenState();
  }
}
class _LoginScreenState extends State<LoginScreen> {
  final emailController = TextEditingController();
  final passwordController = TextEditingController();
  bool isLoading = false;
  bool obscurePassword = true;
  Future<void> login() async {
    final email = emailController.text.trim();
    final password = passwordController.text.trim();
    if (email.isEmpty || password.isEmpty) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(
          content: Text(
            'Please enter email and password',
          ),
        ),
      );
    }
    return;
  }
  setState(() {
    isLoading = true;
  });
  try {
    await FirebaseAuth.instance
      .signInWithEmailAndPassword(
        email: email,
        password: password,

```

```

);
if (!mounted) return;
Navigator.pushReplacement(
  context,
  MaterialPageRoute(
    builder: (context) =>
      const DashboardScreen(),
  ),
);
} on FirebaseAuthException catch (e) {
  String message = 'Login failed';
  if (e.code == 'user-not-found') {
    message = 'No account found with this email';
  }
  else if (e.code == 'wrong-password' ||
    e.code == 'invalid-credential') {
    message = 'Incorrect email or password';
  }
  else if (e.code == 'invalid-email') {
    message = 'Please enter a valid email';
  }
  else if (e.code == 'user-disabled') {
    message = 'This account has been disabled';
  }
  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(
      content: Text(message),
    ),
  );
} catch (e) {
  ScaffoldMessenger.of(context).showSnackBar(
    const SnackBar(
      content: Text(
        'Something went wrong. Please try again.',
      ),
    ),
  );
} finally {
  if (mounted) {
    setState(() {
      isLoading = false;
    });
  }
}

```

```

    }
  }
  @override
  void dispose() {
    emailController.dispose();
    passwordController.dispose();
    super.dispose();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text(
          'Employee Attendance Tracker',
        ),
        centerTitle: true,
      ),
      body: SingleChildScrollView(
        padding: const EdgeInsets.all(20),
        child: Column(
          children: [
            const SizedBox(height: 40),
            const Icon(
              Icons.people,
              size: 90,
              color: Colors.blue,
            ),
            const SizedBox(height: 20),
            const Text(
              'Admin Login',
              style: TextStyle(
                fontSize: 28,
                fontWeight: FontWeight.bold,
              ),
            ),
            const SizedBox(height: 10),
            const Text(
              'Login to manage employee attendance',
              textAlign: TextAlign.center,
              style: TextStyle(
                color: Colors.grey,
                fontSize: 15,
              ),
            ),
          ],
        ),
      ),
    );
  }
}

```

```

),
const SizedBox(height: 40),
TextField(
  controller: emailController,
  keyboardType:
    TextInputType.emailAddress,
  decoration: const InputDecoration(
    labelText: 'Email',
    hintText: 'Enter your email',
    prefixIcon: Icon(
      Icons.email,
    ),
    border: OutlineInputBorder(),
  ),
),
const SizedBox(height: 20),
TextField(
  controller: passwordController,
  obscureText: obscurePassword,
  decoration: InputDecoration(
    labelText: 'Password',
    hintText: 'Enter your password',
    prefixIcon: const Icon(
      Icons.lock,
    ),
    suffixIcon: IconButton(
      icon: Icon(
        obscurePassword
          ? Icons.visibility
          : Icons.visibility_off,
      ),
      onPressed: () {
        setState(() {
          obscurePassword =
            !obscurePassword;
        });
      },
    ),
    border:
      const OutlineInputBorder(),
  ),
),
const SizedBox(height: 30),

```

```

    SizedBox(
      width: double.infinity,
      height: 55,
      child: ElevatedButton.icon(
        onPressed:
          isLoading ? null : login,
        icon: isLoading
          ? const SizedBox(
              width: 20,
              height: 20,
              child:
                CircularProgressIndicator(
                  strokeWidth: 2,
                  color: Colors.white,
                ),
            )
          : const Icon(
              Icons.login,
            ),
        label: Text(
          isLoading
            ? 'Logging in...'
            : 'Login',
        ),
      ),
    ),
    const SizedBox(height: 20),
    const Text (
      'Use the email and password provided by the administrator.',
      textAlign: TextAlign.center,
      style: TextStyle(
        color: Colors.grey,
        fontSize: 13,
      ),
    ),
  ],
),}

```

screens/monthly_reportly_screed.dart:

```

import 'package:flutter/material.dart';
import '../data/app_data.dart';

class MonthlyReportScreen extends StatefulWidget {
  const MonthlyReportScreen({super.key});
  @override
  State<MonthlyReportScreen> createState() =>
    _MonthlyReportScreenState();
}
class _MonthlyReportScreenState
  extends State<MonthlyReportScreen> {
  DateTime selectedMonth = DateTime.now();
  String get monthName {
    const months = [
      'January',
      'February',
      'March',
      'April',
      'May',
      'June',
      'July',
      'August',
      'September',
      'October',
      'November',
      'December',
    ];
    return '${months[selectedMonth.month - 1]} ${selectedMonth.year}';
  }
  bool isSameMonth(String date) {
    final parts = date.split('/');
    if (parts.length != 3) {
      return false;
    }
  }
}

```

```

    final month = int.tryParse(parts[1]);
    final year = int.tryParse(parts[2]);
    return month == selectedMonth.month &&
        year == selectedMonth.year;
}
void changeMonth(int value) {
  setState(() {
    selectedMonth = DateTime(
      selectedMonth.year,
      selectedMonth.month + value,
    );
  });
}
@override
Widget build(BuildContext context) {
  final records = AppData.attendanceRecords
    .where((record) => isSameMonth(record.date))
    .toList();
  final present = records
    .where((record) => record.status == 'Present')
    .length;
  final absent = records
    .where((record) => record.status == 'Absent')
    .length;
  final leave = records
    .where((record) => record.status == 'Leave')
    .length;
  return Scaffold(
    appBar: AppBar(
      title: const Text('Monthly Report'),
      centerTitle: true,
    ),
    body: Padding(
      padding: const EdgeInsets.all(16),
      child: Column(
        children: [
          Row(
            mainAxisAlignment:
              MainAxisAlignment.spaceBetween,
            children: [
              IconButton(
                onPressed: () {
                  changeMonth(-1);

```

```

    },
    icon: const Icon(
      Icons.arrow_back,
    ),
  ),
  Text(
    monthName,
    style: const TextStyle(
      fontSize: 20,
      fontWeight: FontWeight.bold,
    ),
  ),
  IconButton(
    onPressed: () {
      changeMonth(1);
    },
    icon: const Icon(
      Icons.arrow_forward,
    ),
  ),
],
),
const SizedBox(height: 20),
Row(
  children: [
    Expanded(
      child: _reportCard(
        'Present',
        present.toString(),
        Icons.check_circle,
        Colors.green,
      ),
    ),
    const SizedBox(width: 10),
    Expanded(
      child: _reportCard(
        'Absent',
        absent.toString(),
        Icons.cancel,
        Colors.red,
      ),
    ),
    const SizedBox(width: 10),
  ],
)

```

```

Expanded(
  child: _reportCard(
    'Leave',
    leave.toString(),
    Icons.event_busy,
    Colors.orange,
  ),
),
],
),
const SizedBox(height: 25),
Align(
  alignment: Alignment.centerLeft,
  child: Text(
    'Attendance Records',
    style: const TextStyle(
      fontSize: 20,
      fontWeight: FontWeight.bold,
    ),
  ),
),
const SizedBox(height: 10),
Expanded(
  child: records.isEmpty
    ? const Center(
      child: Text(
        'No attendance records found',
        style: TextStyle(
          fontSize: 17,
          color: Colors.grey,
        ),
      ),
    )
    : ListView.builder(
      itemCount: records.length,
      itemBuilder: (context, index) {
        final record = records[index];
        return Card(
          margin: const EdgeInsets.only(
            bottom: 10,
          ),
          child: ListTile(
            leading: CircleAvatar(

```

```

        child: Text(
          record.employeeName[0]
            .toUpperCase(),
        ),
      ),
      title: Text(
        record.employeeName,
        style: const TextStyle(
          fontWeight: FontWeight.bold,
        ),
      ),
      subtitle: Text(
        '${record.employeeId}\n'
        '${record.date}',
      ),
      isThreeLine: true,
      trailing: Text(
        record.status,
        style: TextStyle(
          fontWeight: FontWeight.bold,
          color:
            record.status == 'Present'
              ? Colors.green
              : record.status ==
                'Absent'
              ? Colors.red
              : Colors.orange,
        ),
      ),
    ),
  );
},
),
],
),
),
);
}
Widget _reportCard(
  String title,
  String value,
  IconData icon,

```

```

    Color color,
  ) {
return Card(
  child: Padding(
    padding: const EdgeInsets.all(12),
    child: Column(
      children: [
        Icon(
          icon,
          color: color,
          size: 28,
        ),
        const SizedBox(height: 6),
        Text(
          value,
          style: const TextStyle(
            fontSize: 22,
            fontWeight: FontWeight.bold,
          ),
        ),
        Text(title),
      ],
    ),
  ),
);
}
}

```

screens/splash_screen.dart:

```

import 'dart:async';
import 'package:flutter/material.dart';
import 'login_screen.dart';

class SplashScreen extends StatefulWidget {
  const SplashScreen({
    super.key,
  });
  @override
  State<SplashScreen> createState() =>
    _SplashScreenState();
}
class _SplashScreenState
  extends State<SplashScreen> {
  @override
  void initState()
  super.initState();
  Timer(
    const Duration(seconds: 3),
    () {
      if (!mounted) return;
      Navigator.pushReplacement(
        context,
        MaterialPageRoute(
          builder: (context) =>
            const LoginScreen(),
        ),
      );
    },
  );
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Center(
      child: Column(
        mainAxisAlignment:

```

```

MainAxisAlignment.center,
children: [
  const Icon(
    Icons.people,
    size: 90,
    color: Colors.blue,
  ),
  const SizedBox(height: 20),
  const Text(
    'Employee Attendance Tracker',
    style: TextStyle(
      fontSize: 24,
      fontWeight: FontWeight.bold,
    ),
    textAlign: TextAlign.center,
  ),
  const SizedBox(height: 10),
  const Text(
    'Manage employee attendance easily',
    style: TextStyle(
      fontSize: 15,
      color: Colors.grey,
    ),
  ),
  const SizedBox(height: 30),
  const CircularProgressIndicator(),
],
),
),
);
}
}

```

widgets/firebase_options.dart:

```
import 'package:firebase_core/firebase_core.dart';
class DefaultFirebaseOptions{
  static FirebaseOptions get currentPlatform{
    return const FirebaseOptions(
      apiKey: "AIzaSyBuCevJhoRVlxppgzcqPPcU6MuRG-DqvL0",
      authDomain: "employee-attendance-trac-5d736.firebaseio.com",
      projectId: "employee-attendance-trac-5d736",
      storageBucket: "employee-attendance-trac-5d736.firebaseio.com",
      messagingSenderId: "421981021134",
      appId: "1:421981021134:web:b3727b4781855fa78e419f"
    );
  }
}
```

widgets/main.dart:

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'firebase_options.dart';
import 'screens/splash_screen.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp(
    options:
      DefaultFirebaseOptions.currentPlatform,
  );
  runApp(
    const EmployeeAttendanceApp(),
  );
}

class EmployeeAttendanceApp
  extends StatelessWidget {
  const EmployeeAttendanceApp({
    super.key,
```

```

});

@override
Widget build(BuildContext context) {
  return MaterialApp(
    debugShowCheckedModeBanner: false,
    title:
    'Employee Attendance Tracker',
    theme: ThemeData(
      primarySwatch: Colors.blue,
      useMaterial3: true,
    ),
    home:
    const SplashScreen(),
  );
}
}

```

test/widget_test.dart:

```

import 'package:flutter_test/flutter_test.dart';
void main() {
  test('Employee Attendance Tracker test', () {
    expect(1 + 1, 2);
  });
}

```

pubspec.yaml:

```

dependencies:
  flutter:
    sdk: flutter
  firebase_core: ^4.1.0
  firebase_auth: ^6.0.2

```

Output:**Authentication page:**

Employee Attendance Tracker



Admin Login
Login to manage employee attendance

Email

Password

Login

Use the email and password provided by the administrator.

Incorrect email or password message:

Employee Attendance Tracker



Admin Login
Login to manage employee attendance

Email

Password

Login

Use the email and password provided by the administrator.

Incorrect email or password

Enter the correct email Id and password:

Employee Attendance Tracker



Admin Login

Login to manage employee attendance

Email

 admin@gmail.com

Password

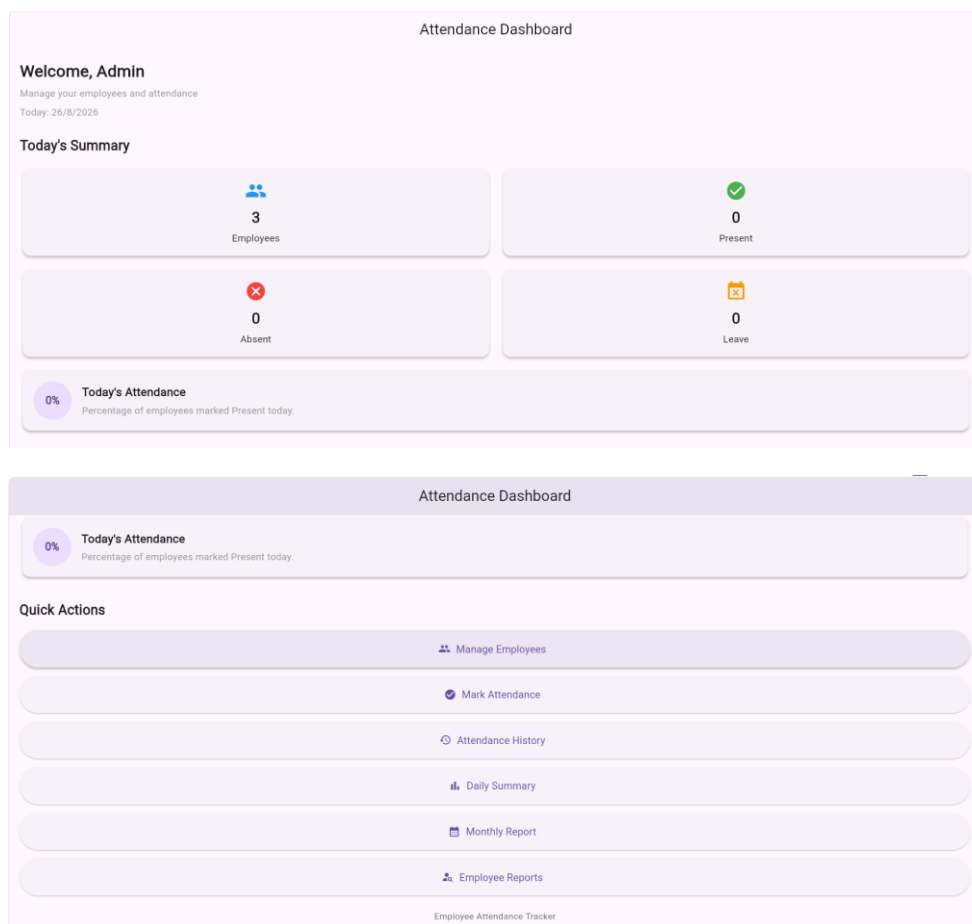
 Admin@123 

 Login

Use the email and password provided by the administrator.

After Logging with email id and password:

Attendance Dashboard:



Manage Employ

 Employees

R

Rahul Sharma
EMP001
IT • Developer
9876543210



A

Aman Khan
EMP002
HR • Manager
9876543211



P

Priya Patel
EMP003
Finance • Accountant
9876543212

 Add Employee

Click on the ‘Add Employee’ Button’:

 Add Employee

 Employee Name

 Department

 Designation

 Phone

 Email

 Joining Date



The phone number should contain exactly 10-digits:

Employee Name

 Harsh Gupta

Department

 Security

Designation

 Bouncer

Phone

 987456321047282

Enter a valid 10-digit phone number

Email

 harshguptagmailcom

Enter a valid email address

Joining Date

 24/8/2026



←

Add Employee

Employee Name

Harsh Gupta

Department

Security

Designation

Bouncer

Phone

9874563210

Email

harshguptagmailcom

Enter a valid email address

Joining Date

24/8/2026

Save Employee

The email ID should contain '@' & '.':

Phone

9874563210

Email

harshgupta@gmailcom

Enter a valid email address

Joining Date

24/8/2026

Now add '@' & '.' and click on 'Save Employee':

Phone

9874563210

Email

harshgupta@gmail.com

Joining Date

24/8/2026

'Employee added successfully' message:

←

Employees

R

Rahul Sharma

EMP001

IT • Developer

9876543210

A

Aman Khan

EMP002

HR • Manager

9876543211

P

Priya Patel

EMP003

Finance • Accountant

9876543212

H

Harsh Gupta

EMP004

Security • Bouncer

9874563210

Add Employee

Employee added successfully

Click 'Edit' button on Employee':

H

Harsh Gupta
EMP004
Security • Bouncer
9874563210

Edit Employee

Add Employee

←

Edit Employee

Employee Name

Harsh Gupta

Department

Security

Designation

Bouncer

Phone

9874563210

Email

harshgupta@gmail.com

Joining Date

24/8/2026

Update Employee

Change the Designation 'Bouncer' to 'Head' & click on 'Update Employee':

Employee Name

Harsh Gupta

Department

Security

Designation

Head

Phone

9874563210

Email

harshgupta@gmail.com

Joining Date

24/8/2026

Update Employee

H

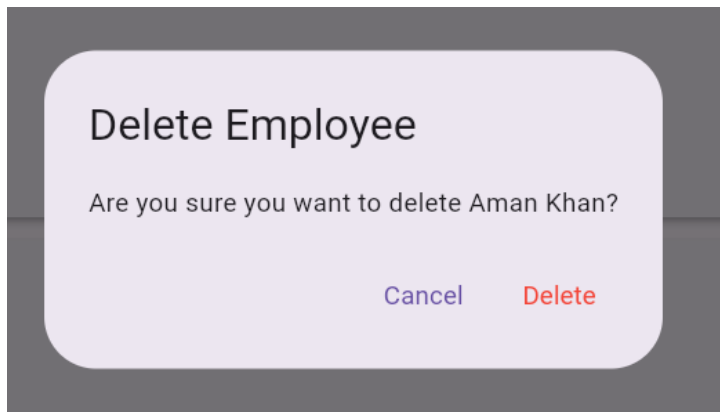
Harsh Gupta
EMP004
Security • Head
9874563210

'Delete' the employee 'Aman Khan':

A

Aman Khan
EMP002
HR • Manager
9876543211

Delete Employee



After 'Delete' Aman khan:

R	Rahul Sharma EMP001 IT • Developer 9876543210	 
P	Priya Patel EMP003 Finance • Accountant 9876543212	 
H	Harsh Gupta EMP004 Security • Head 9874563210	 

Mark Attendance:

 Manage Employees
 Mark Attendance
 Attendance History

Dashboard of ‘Mark Attendance’:

←

Mark Attendance

Mark Employee Attendance
Select employee, date and attendance status.

Select Employee

Employee

Choose employee

Attendance Date

Date

26/8/2026

Attendance Status

☒ Present

☐ Absent

☐ Leave

Save Attendance

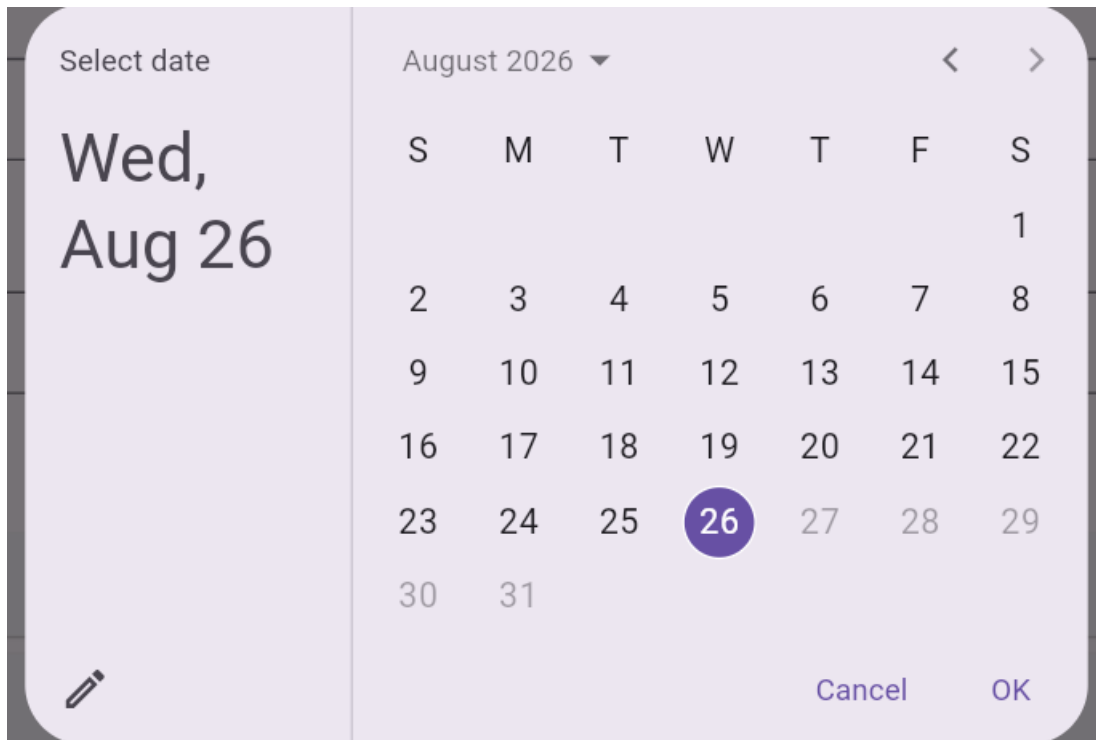
Choose the employee, attendance date, and attendance status:

Select Employee

EMP001 - Rahul Sharma

EMP003 - Priya Patel

EMP004 - Harsh Gupta

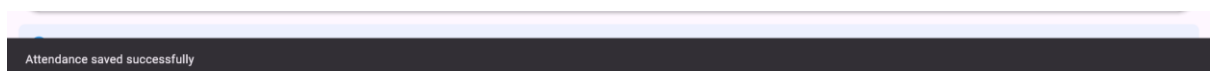


Mark Rahul as absent and click on ‘Save Attendance’:

Attendance Status

☐ Present	✓
☒ Absent	✗
☐ Leave	📅

After click on ‘Save Attendance’:



When I try to mark 'Rahul' as 'Present', it shows 'Attendance already marked'.

Select Employee

Employee

 EMP001 - Rahul Sharma

Attendance Date

Date

 26/8/2026

Attendance Status

☒ Present

☐ Absent

☐ Leave

Attendance already marked for Rahul Sharma on 26/8/2026

Attendance History:

Here, the admin can change the attendance status:

Attendance History	
Back	
R Rahul Sharma EMP001 26/8/2026	Absent
A Aman Khan EMP002 26/8/2026	Present
P Priya Patel EMP003 26/8/2026	Leave
H Harsh Gupta EMP004 26/8/2026	Present

Click on 'Edit Attendance' icon:

P Priya Patel EMP003 26/8/2026	Leave Edit Attendance
--	---

Change the attendance status from 'Leave' to 'Present' and click on the 'Update' button:

Edit Attendance

Employee

EMP003 - Priya Patel

Date

26/8/2026

Attendance Status

☐

 Present

☐

 Absent

☒

 Leave

Cancel

Update

Edit Attendance

Employee

EMP003 - Priya Patel

Date

26/8/2026

Attendance Status

☒

 Present

☐

 Absent☐

Cancel

Update

‘Attendance updated successfully’:

P	Priya Patel EMP003 26/8/2026	Present  
H	Harsh Gupta EMP004 26/8/2026	Present  

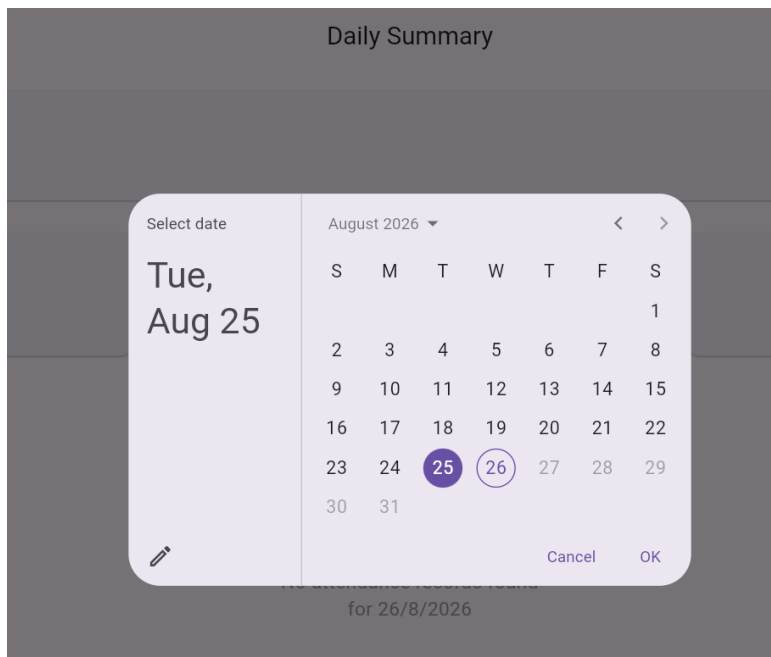
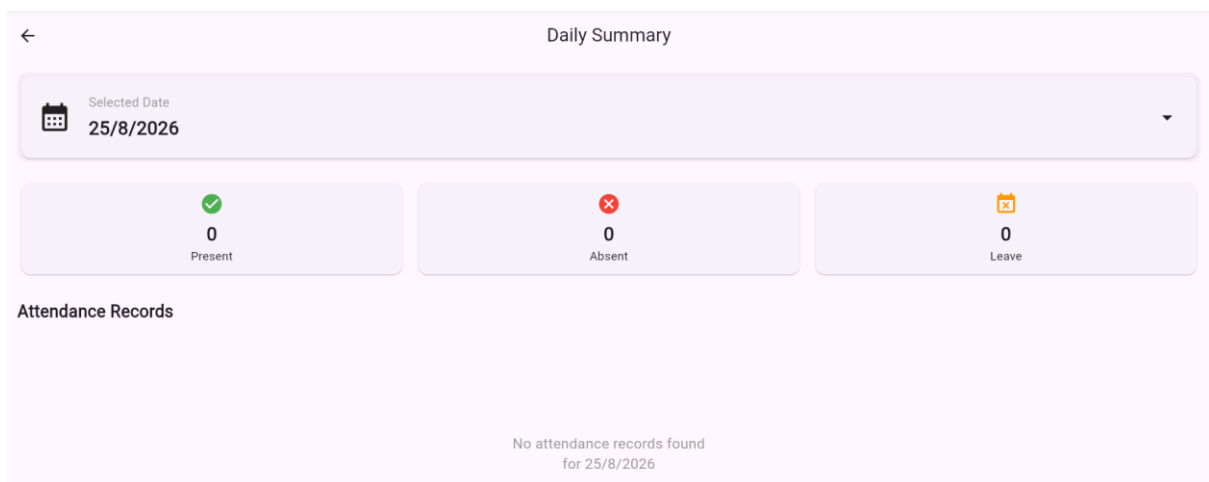
Attendance updated successfully

Daily Summary:

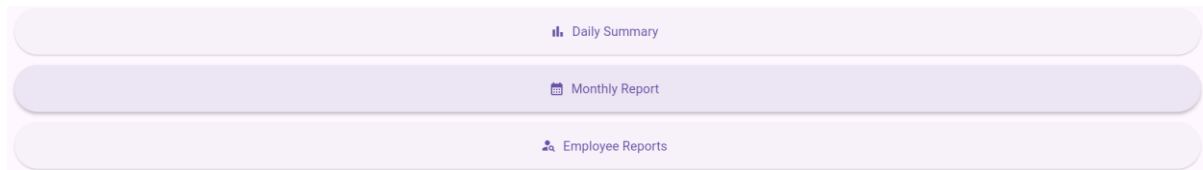
 Attendance History
 Daily Summary

Here, we can view date-wise attendance records:

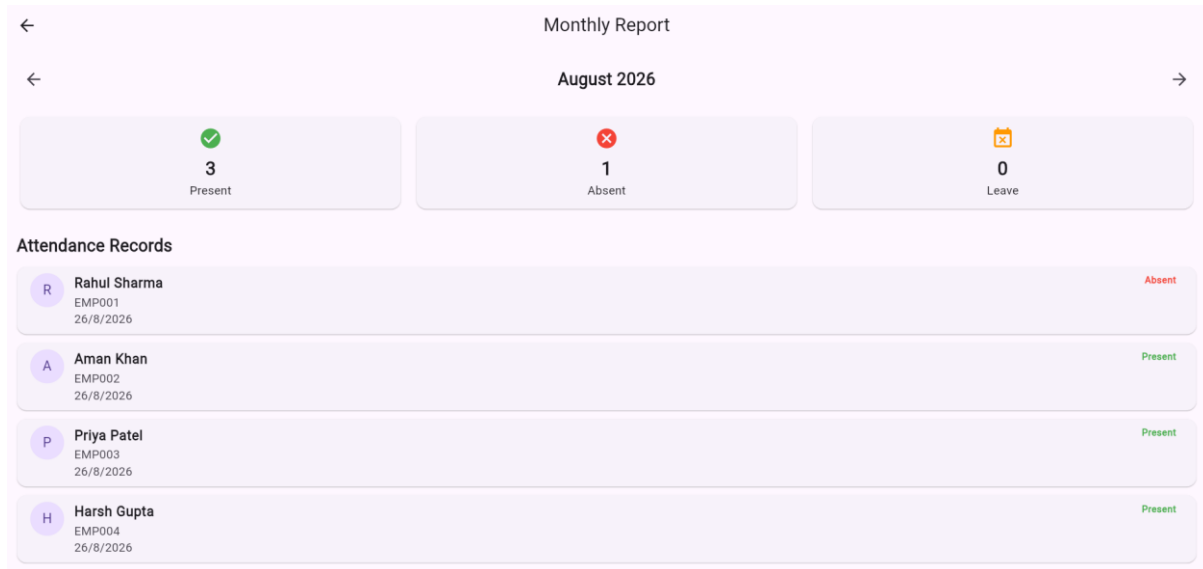
Daily Summary		
Today 26/8/2026		
 3 Present	 1 Absent	 0 Leave
Today's Attendance Records		
R	Rahul Sharma EMP001	Absent
A	Aman Khan EMP002	Present
P	Priya Patel EMP003	Present
H	Harsh Gupta EMP004	Present

Change the date and click on ‘OK’ Button:**No record found:**

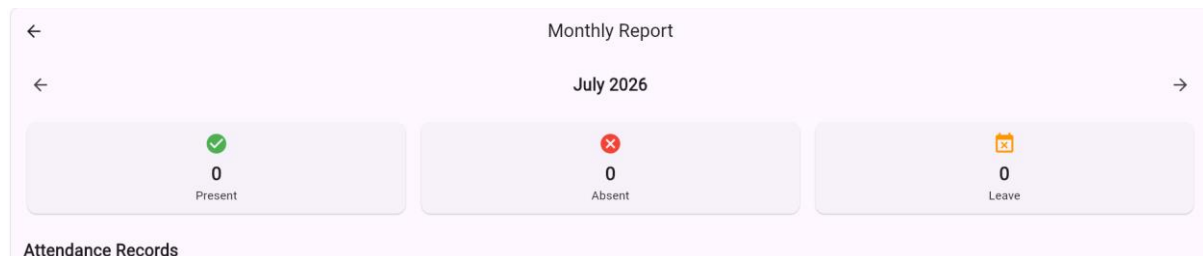
Monthly Report:



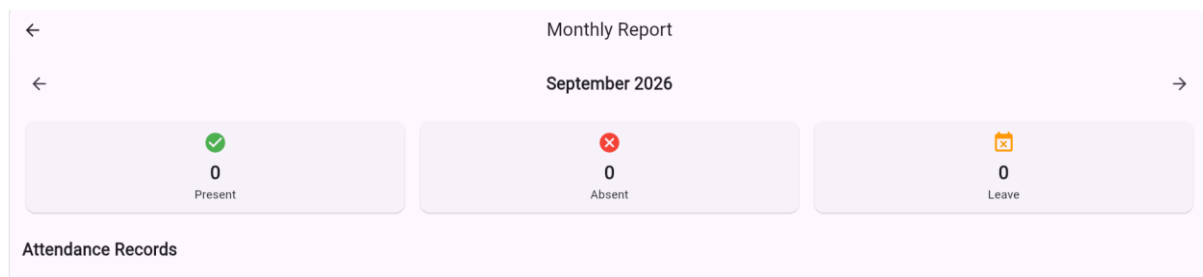
Here, we can view the monthly attendance report of employee:



Attendance report for July:

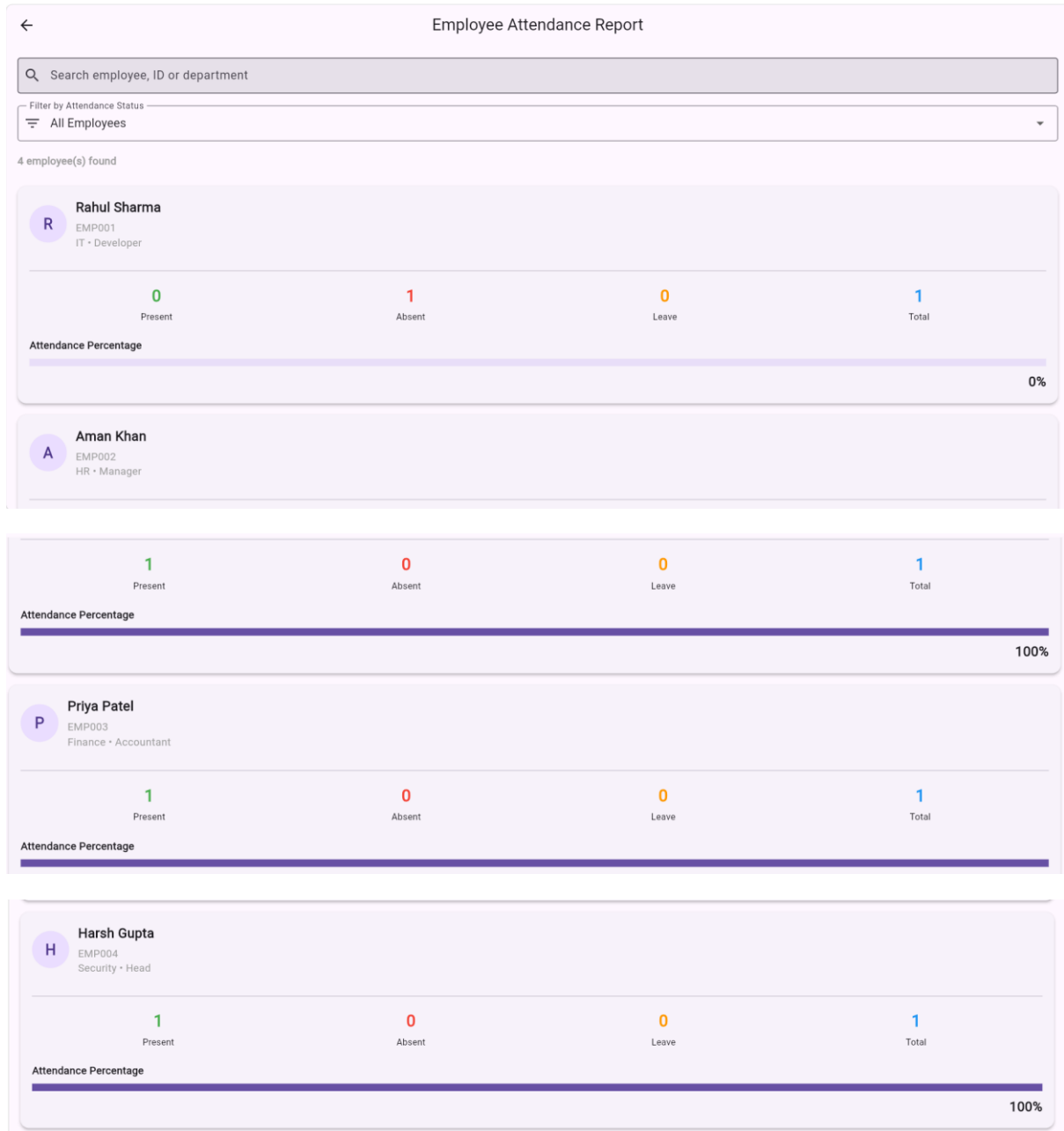


Attendance report for September:



Employee Reports:

Employee Reports dashboard:



Filter by employees marked as Present.

←

Employee Attendance Report

All Employees

Present

Absent

Leave

3 employees found who are present:

Filter by Attendance Status

Present

3 employee(s) found

A

Aman Khan

EMP002

HR • Manager

1

Present

0

Absent

0

Leave

1

Total

Attendance Percentage

100%

P

Priya Patel

EMP003

Finance • Accountant

1

Present

0

Absent

0

Leave

1

Total

Attendance Percentage

100%

H

Harsh Gupta

EMP004

Security • Head

1

Present

0

Absent

0

Leave

1

Total

Attendance Percentage

100%

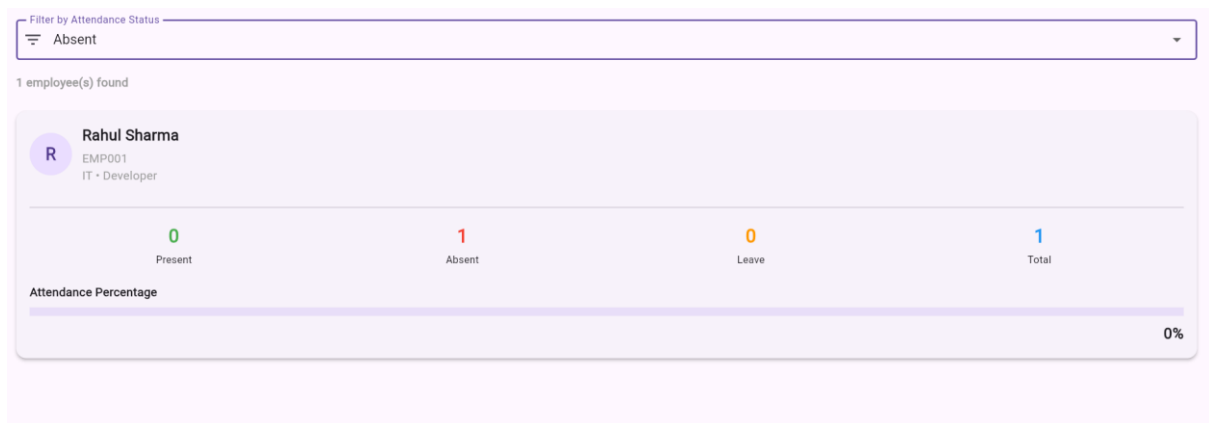
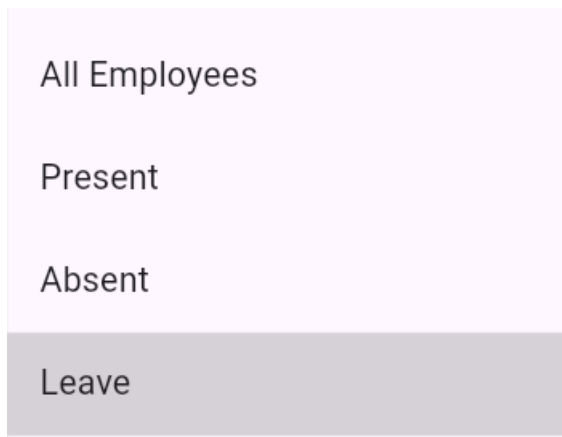
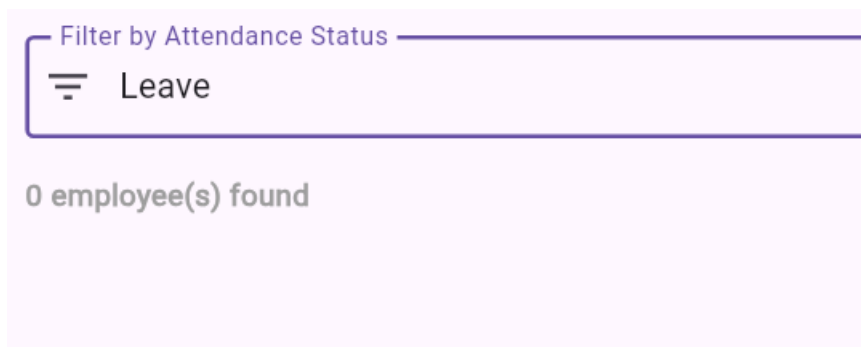
Filter by employees marked as Absent:

All Employees

Present

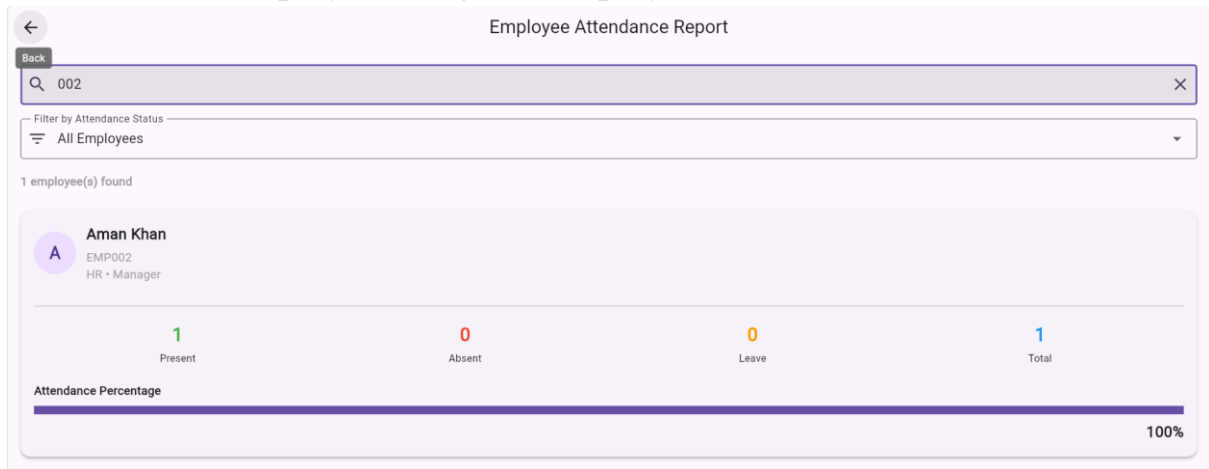
Absent

Leave

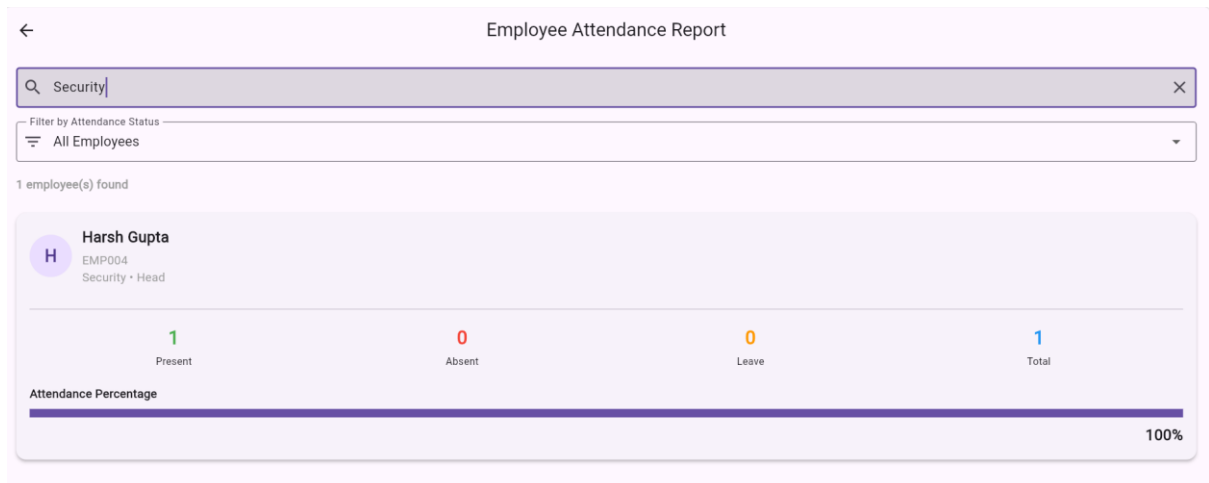
1 employee was found with an Absent status:**Filter by 'Leave':****0 employees found:**

Search Bar:

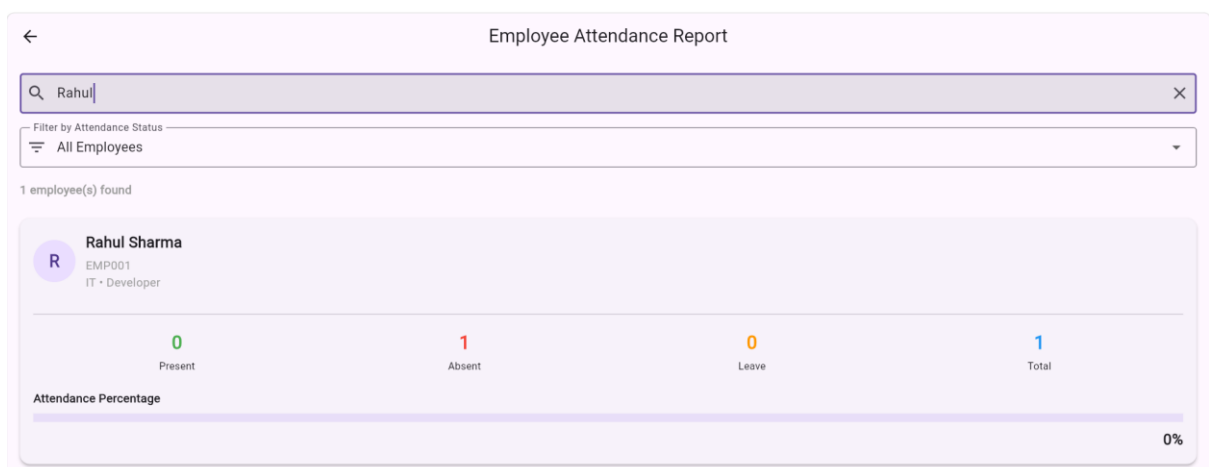
Search for an employee using the Employee ID:



Search for an employee using the department:



Search for an employee using the employee name:



Updated dashboard after marking employee attendance:**Total: 4 | Present: 3 | Absent: 1 | Leave: 0 | Today's Attendance: 75%**